PROJECT DOCUMENTATION

Smart Library Request Workflow in ServiceNow

— Technical Project Report —

1. Introduction

Documentation constitutes a fundamental pillar of any well-engineered information system project. In the context of the Smart Library Request Workflow developed on the ServiceNow platform, comprehensive documentation serves as the authoritative reference that captures every decision, configuration, and process involved in the system's lifecycle. From the initial identification of requirements through to deployment and user training, documentation ensures that the project remains traceable, reproducible, and maintainable across time and organisational change.

The importance of structured documentation cannot be overstated. It promotes clarity by providing unambiguous descriptions of system behaviour and configuration, thereby reducing the risk of misinterpretation by stakeholders. It facilitates knowledge transfer by enabling new team members, administrators, or auditors to understand the system without reliance on individuals who participated in its original development. Furthermore, it supports future maintenance by supplying the contextual detail necessary for system administrators to diagnose issues, implement enhancements, or migrate the system to alternative environments.

This documentation section provides a structured and comprehensive account of all artefacts produced throughout the project's duration, organised by phase and functional domain. Each subsection addresses a distinct dimension of the system and follows professional technical writing standards appropriate for academic project reporting.

2. Documentation Overview

The documentation for the Smart Library Request Workflow project spans the entirety of the project lifecycle, covering all six principal phases from inception through to deployment and user support. Collectively, these documents form a complete and authoritative guide that equips stakeholders with the knowledge required to understand, operate, and recreate the system in its entirety.

2.1 Project Phases Covered

The documentation encompasses the following project phases, each of which is addressed with dedicated artefacts, diagrams, and descriptive records:

Identification Phase

This phase documents the problem statement, project scope, stakeholder analysis, and the rationale for selecting ServiceNow as the implementation platform. It establishes the foundational context for all subsequent project activities.

Requirement Analysis Phase

Documentation from this phase includes functional and non-functional requirements, use case descriptions, and stakeholder interviews. These records define precisely what the system must achieve to be considered successful.

Project Planning Phase

Planning documentation covers the project schedule, resource allocation, risk assessment matrix, and milestone definitions. This phase ensures that the project's execution strategy is clearly articulated and measurable.

Project Design Phase

Design artefacts include system architecture diagrams, entity-relationship models, workflow flowcharts, and user interface wireframes. These documents translate requirements into a blueprint for technical implementation.

Project Development Phase

Development documentation records the configuration steps carried out within ServiceNow, including table creation, field definitions, role assignments, and automation scripts. This phase provides a step-by-step recreation guide for system administrators.

Testing and Deployment Phase

This phase documents test cases, test results, defect logs, and the deployment procedure. It validates that the system operates as intended and that all acceptance criteria have been satisfied prior to go-live.

Taken together, these phase-specific documents provide a complete and coherent guide that any qualified system administrator or developer could use to fully understand and independently recreate the Smart Library Request Workflow system within a ServiceNow environment.

3. System Setup Documentation

System setup documentation provides a precise account of all configuration steps required to establish the Smart Library Request Workflow within the ServiceNow platform. This documentation is critical for system administrators responsible for maintaining the production environment or replicating the configuration in development and testing instances. The following subsections describe each configuration domain in detail.

3.1 Role Creation

The system operates under a two-role security model, comprising the Student role and the Librarian role. The Student role is designed to grant registered library users the ability to submit borrow requests and monitor the status of their submissions. The Librarian role is assigned to library staff and provides elevated privileges, including the ability to review pending requests, approve or reject submissions, manage the book inventory, and update book availability status.

Role creation documentation records the role names, their associated permissions, the groups to which they are assigned, and any inheritance relationships between roles. This ensures that access control configurations can be reproduced accurately and audited against security policy requirements.

3.2 Table Creation

Two custom tables were created within ServiceNow to underpin the system's data architecture. The Book table serves as the central catalogue, storing records for all library holdings including title, author, ISBN, availability status, and category. The Borrow Request table manages transactional records relating to borrowing activity, capturing details such as the requesting student, the requested book, submission date, approval status, due date, and any librarian remarks.

Table documentation specifies the table name, label, parent table (where applicable), access controls, and a complete field inventory. This level of detail ensures that the relational structure of the system is fully transparent and reproducible.

3.3 Field Configuration

Field configuration documentation describes each field defined within both custom tables, including its data type, label, internal name, default value, mandatory status, and display order. Special attention is given to reference fields, which establish relational links between records, and to choice fields, which constrain user input to predefined values such as "Available," "Borrowed," "Pending," "Approved," and "Rejected."

3.4 Reference Qualifier Setup

Reference qualifiers are documented to explain how the system restricts the selectable values in reference fields. For instance, the book reference field on the Borrow Request form is configured with a qualifier that limits the selectable books to those with an availability status of "Available." This prevents students from submitting requests for books that are already on loan. The documentation records the qualifier logic, its placement, and the conditions under which it is applied.

Together, these configuration components form the structural foundation of the system. Without accurate documentation of each element, the system would be difficult to audit, troubleshoot, or replicate, making this section among the most critical within the broader project record.

4. Workflow and Automation Documentation

Workflow and automation documentation captures the logic, configuration, and behaviour of the automated processes that govern the library borrowing lifecycle. These processes were implemented using ServiceNow's Flow Designer, a visual automation tool that enables the creation of conditional, multi-step workflows without requiring custom scripting. The documentation provides both a high-level process narrative and granular technical specifications for each automation component.

4.1 Borrow Request Submission

The workflow is triggered automatically upon the submission of a new Borrow Request record. Documentation for this stage records the trigger condition, the initiating event type, and the initial state transitions applied to both the request record and the associated book record. The process

ensures that upon submission, the request status is set to “Pending” and the selected book’s availability is updated to reflect that it is under consideration for loan.

4.2 Approval Process by Librarians

The approval stage documentation describes the conditions under which a librarian is notified of a pending request, the interface through which they review and act upon it, and the branching logic that differentiates between approval and rejection outcomes. Each outcome triggers a distinct subsequent action, ensuring that the system responds appropriately to the librarian’s decision without requiring manual intervention beyond the approval action itself.

4.3 Automatic Book Status Updates

Automation documentation records how the system updates the Book table record automatically in response to workflow events. Upon approval, the relevant book’s status field is updated to “Borrowed.” Upon rejection or return, the status reverts to “Available.” These automated field updates are documented with reference to the specific Flow Designer actions responsible, including any conditions or filters applied to ensure they execute only under the correct circumstances.

4.4 Email Notifications to Users

The notification documentation specifies each automated email that the system generates throughout the borrowing process. Notifications are sent to students upon submission confirmation, approval, and rejection of their requests. Librarians receive notifications when new requests are submitted and when overdue books are identified. Each notification record includes the email template name, the trigger condition, the recipient logic, and a sample of the email body to illustrate the information communicated to users.

5. User Interface Documentation

User interface documentation describes the layout, behaviour, and configuration of the forms, views, and related lists presented to system users within the ServiceNow portal and backend interface. This documentation is essential for understanding how users interact with the system and for ensuring that interface behaviour aligns with the business requirements established during the analysis phase.

5.1 Borrow Request Form Behaviour

The Borrow Request form documentation details the fields presented to students when submitting a new borrowing request, including their order, labels, data types, and mandatory status. It records the dynamic behaviour of the form, such as fields that become visible or mandatory in response to user input, and describes how the reference qualifier on the book field restricts the available selection to currently available titles. Annotated screenshots supplement the written descriptions to provide a clear visual reference for system trainers and support personnel.

5.2 Book Management Interface

The book management interface documentation describes the list view and form view configurations available to librarians for managing the book catalogue. It specifies which fields are displayed in the

list view, the sorting and filtering options available, and the full set of fields accessible on the book record form. This documentation assists librarians in performing routine catalogue maintenance tasks, including adding new titles, updating availability, and retiring outdated records.

5.3 UI Policy Rules

UI policy documentation records the rules that dynamically control form behaviour based on record state or user input. These policies enforce mandatory field requirements, restrict read-only fields to prevent unauthorised modification, and conditionally hide or display form sections based on workflow stage. For example, the rejection reason field is documented as hidden until the librarian selects a rejection outcome, at which point it becomes mandatory. Each UI policy is documented with its name, trigger condition, and the specific field actions it produces.

6. Security and Access Control Documentation

Security and access control documentation describes the Access Control Rules (ACLs) configured within ServiceNow to enforce appropriate permission boundaries for each user role. This documentation is essential for demonstrating that the system adheres to the principle of least privilege, whereby each user is granted only the minimum level of access required to fulfil their functional responsibilities.

6.1 Student Access Permissions

Students are granted the ability to create new Borrow Request records and to view the records they have personally submitted. Their access is restricted to their own data; they are not permitted to view requests submitted by other students, nor can they access book management functions or approval interfaces. ACL documentation for this role specifies the applicable table, the operation type (create, read, write, delete), and the scripted conditions that enforce record-level filtering to the student's own user reference.

6.2 Librarian Access Permissions

Librarians are granted comprehensive management access across both the Book table and the Borrow Request table. This includes the ability to create, read, update, and delete book records, as well as to review all pending requests and execute approval or rejection actions. ACL documentation for the Librarian role records each permission explicitly, including any field-level restrictions that prevent librarians from modifying system-generated or audit fields. This documentation also describes the administrative interfaces available to library staff and the manner in which elevated permissions are assigned through group membership.

7. Reporting Documentation

Reporting documentation describes the analytical reports created within ServiceNow to support data-driven decision-making by library administrators. These reports aggregate transactional data from the Book and Borrow Request tables to surface meaningful insights regarding system usage, collection demand, and operational efficiency.

7.1 Most Borrowed Books Report

The “Most Borrowed Books” report is designed to identify the titles with the highest borrowing frequency over a configurable time period. Documentation for this report records the report type, the data source table, the grouping field, the aggregation method, the filter conditions, and the visualisation format selected. This report enables librarians to make informed decisions regarding collection prioritisation, procurement of additional copies, and the retirement of underutilised titles.

7.2 Request Tracking and Analytics

Request tracking reports provide a real-time and historical view of the volume, status distribution, and processing times associated with borrow requests. Documentation for these reports includes the metrics captured (such as average approval time and rejection rate), the dashboard widgets in which they are embedded, and the intended audience for each report. These analytics support library management in monitoring operational performance and identifying bottlenecks within the approval workflow.

8. Testing Documentation

Testing documentation provides a complete record of the validation activities conducted to verify that the Smart Library Request Workflow operates in accordance with its specified requirements. It encompasses test plans, individual test cases, execution records, and outcome summaries, thereby establishing a transparent and auditable quality assurance trail.

8.1 Workflow Testing

Workflow testing documentation records the test scenarios used to validate the end-to-end operation of the Flow Designer automation. Each test case specifies the input conditions, the expected system response, and the actual outcome observed during execution. Tests cover the primary approval path, the rejection path, edge cases such as the submission of a request for a book already on loan, and the correct triggering of email notifications at each workflow stage.

8.2 Role-Based Access Testing

Role-based access testing validates that the ACL configurations function correctly for both the Student and Librarian roles. Documentation for this test category records scenarios in which test accounts assigned to each role attempt to perform operations both within and beyond their permitted scope. Results confirm that students are unable to access librarian-specific interfaces and that librarians can perform all required administrative operations without encountering permission errors.

8.3 Form Validation Checks

Form validation testing confirms that the UI policies and mandatory field configurations operate as intended. Test cases document scenarios in which users attempt to submit forms with missing required fields, select unavailable books through the reference field, or bypass read-only restrictions on protected fields. Documentation records the validation messages displayed to users and confirms that the system prevents invalid submissions in all tested scenarios.

8.4 Report Accuracy Verification

Report accuracy testing ensures that the data presented in system-generated reports corresponds correctly to the underlying transaction records. Test scenarios involve the creation of controlled test data sets, the execution of reports, and the manual cross-referencing of report outputs against source records. Documentation records the data sets used, the reports tested, and the degree of correspondence observed, providing assurance that analytics outputs can be relied upon for operational decision-making.

9. User Guide and Support

User guidance documentation is a dedicated component of the project record designed to support end users in interacting effectively with the Smart Library Request Workflow system. Unlike the technical documentation sections, which target system administrators and developers, the user guide is written in accessible, task-oriented language suitable for individuals with no prior ServiceNow experience.

9.1 Student User Guide

The student user guide provides step-by-step instructions for accessing the library portal, searching the available book catalogue, submitting a borrow request, and monitoring the status of submitted requests. Each procedural section is accompanied by annotated screenshots that correspond to the actual system interface, enabling students to follow the guidance without prior training. The guide also includes a frequently asked questions section addressing common scenarios such as what to do when a desired book is unavailable or how to contact library support.

9.2 Librarian User Guide

The librarian user guide covers the operational procedures required to manage the day-to-day functions of the library system. It provides guidance on reviewing and processing pending borrow requests, updating book availability records, adding new titles to the catalogue, and generating management reports. Instructions are presented in a sequential format aligned with the librarian's typical daily workflow, with cross-references to relevant system policies and approval procedures where applicable.

9.3 Support Resources

The documentation includes references to available support channels, including the designated system administrator contact, the internal ServiceNow help portal, and the escalation procedure for unresolved technical issues. This ensures that users encountering difficulties beyond the scope of the user guide have a clear and well-documented path to resolution, minimising disruption to library operations.

10. Documentation Outcome

The comprehensive documentation produced throughout the Smart Library Request Workflow project delivers a set of outcomes that extend well beyond the immediate project timeline. These outcomes

collectively ensure that the system remains transparent, maintainable, and scalable as organisational needs evolve.

10.1 System Transparency

By providing explicit records of every configuration decision, workflow logic, and access control rule, the documentation eliminates ambiguity regarding system behaviour. Stakeholders at all levels of the organisation, including library management, IT administrators, and institutional governance bodies, can consult the documentation to obtain an accurate and complete understanding of how the system operates and why it was designed in its current form.

10.2 Easier Maintenance

Maintenance activities, whether involving bug resolution, configuration adjustments, or the onboarding of new administrative staff, are significantly simplified when supported by thorough documentation. System administrators can quickly locate the relevant configuration records, understand the intended behaviour of each component, and make targeted changes with confidence that they are not inadvertently disrupting interdependent system elements.

10.3 Future Scalability

As the library's operational requirements grow, the system documentation provides the foundation necessary for informed and coherent expansion. Future enhancements, such as the introduction of reservation queuing, integration with external catalogue systems, or the extension of the workflow to support digital resource lending, can be designed with full awareness of the existing architecture, thereby reducing the risk of conflicting configurations or duplicated effort.

In summary, the documentation produced for the Smart Library Request Workflow project represents a significant and enduring contribution to the system's long-term viability. It embeds the institutional knowledge generated during the project into a structured, accessible, and reusable reference corpus that will continue to support the effective operation and development of the system for years to come.

End of Project Documentation Section

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